# Development of Face

**Department of Oral Biology**

**Dow University of Health Sciences, Karachi**

# TASK: IDENTIFY, STUDY AND PASTE THE DIAGRAMS

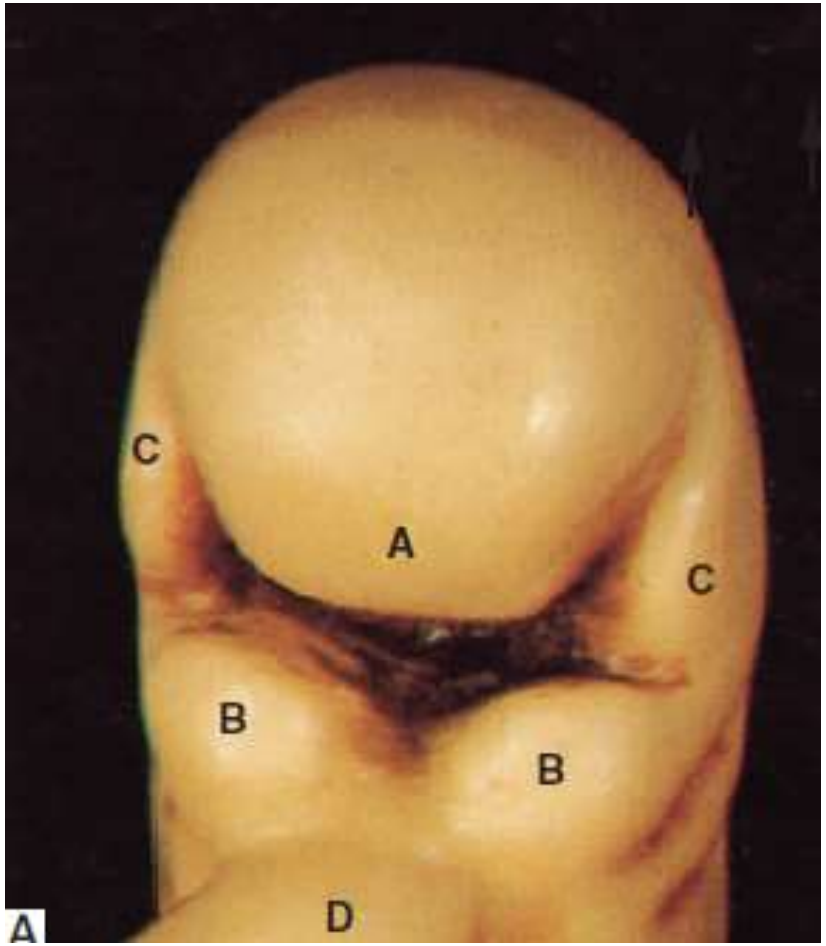



Fig. 17.1.(A) Model of a face of a 4-week-old human embryo

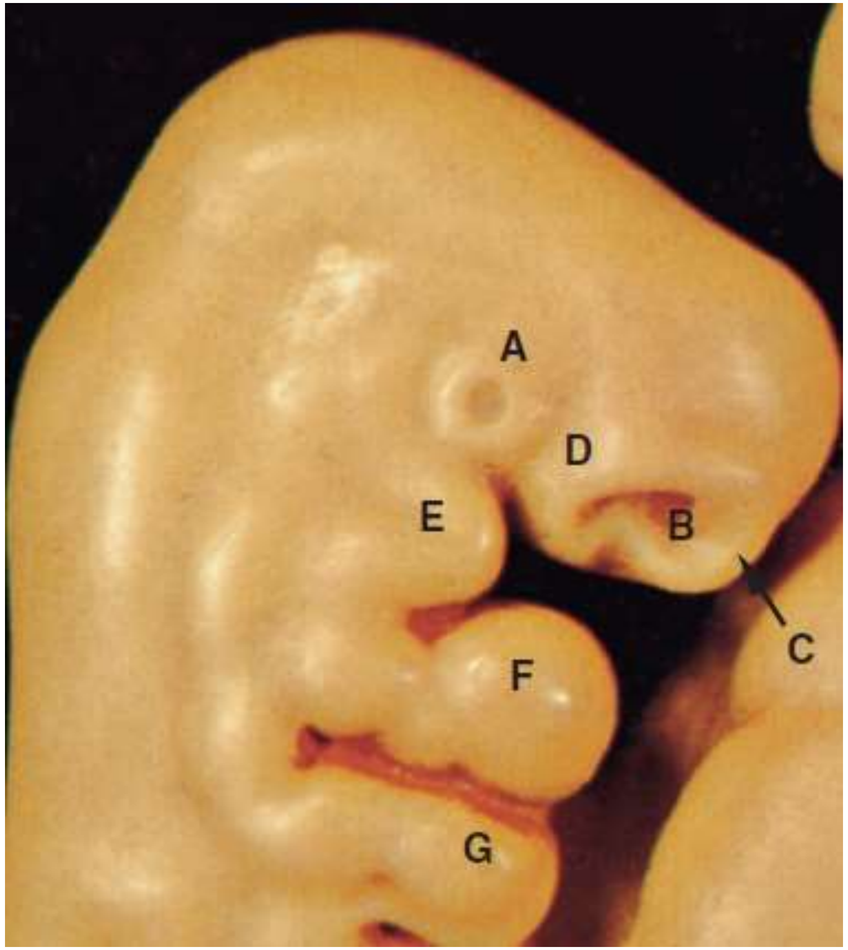



Fig. 17.2 (A) Model of a face of a 5-week-old human embryo

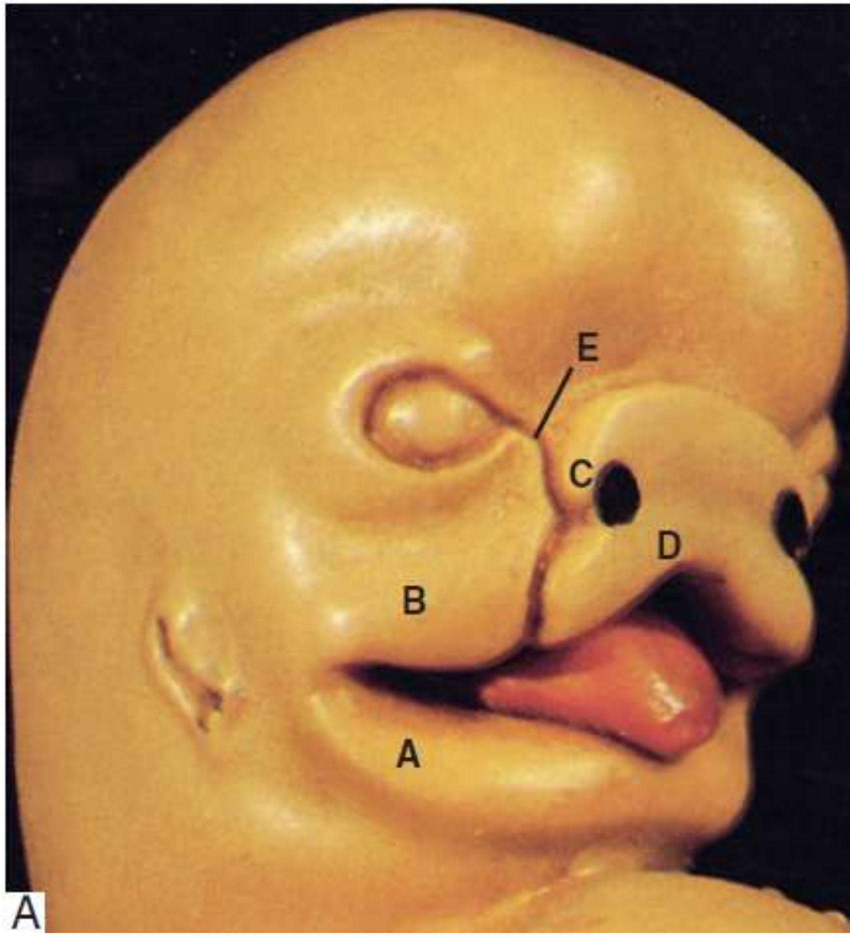
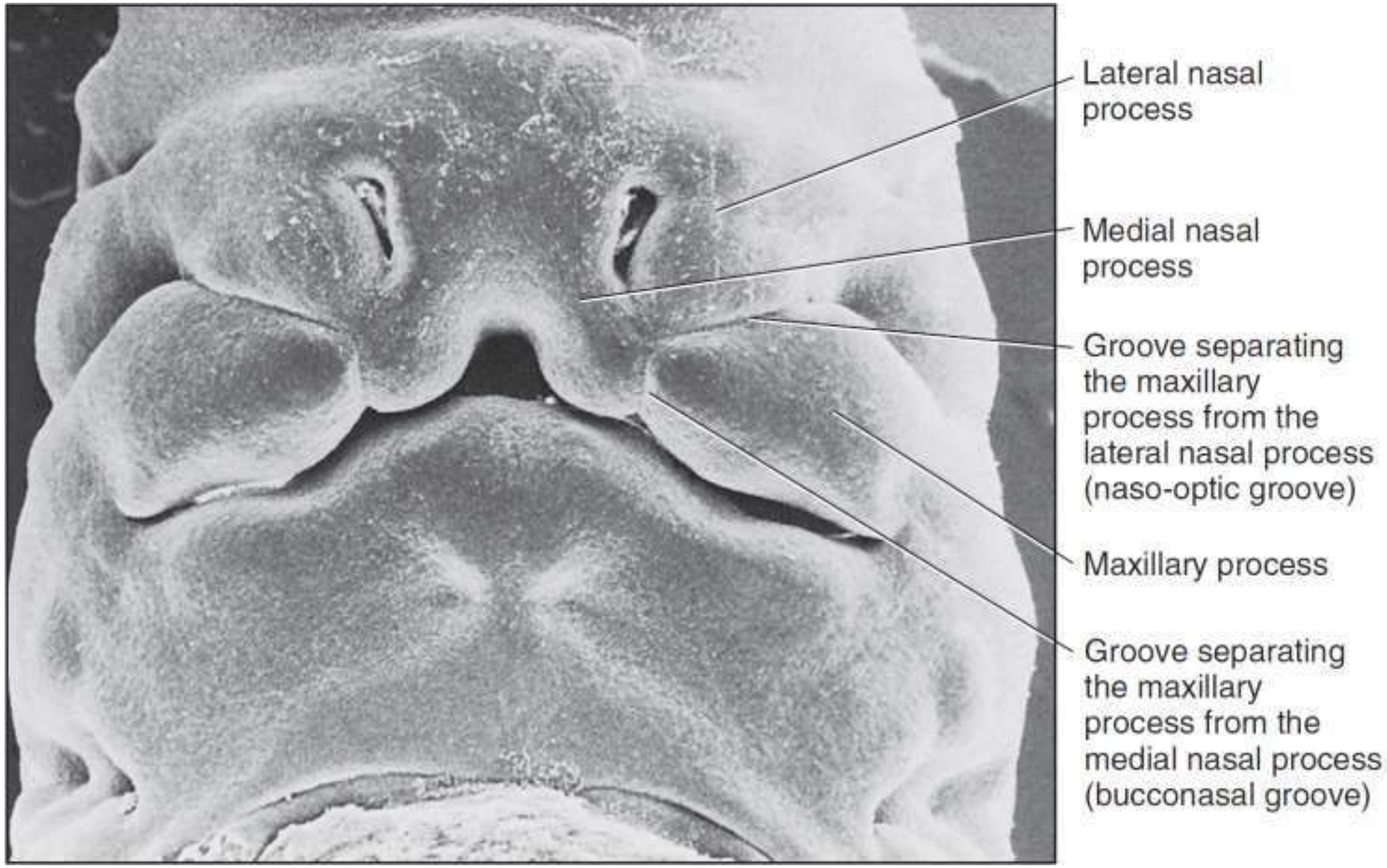



Fig. 17.3 (A) Model of a face of a 6-week-old human embryo




**Scanning electron micrograph of a human embryo at around 6 weeks of development**

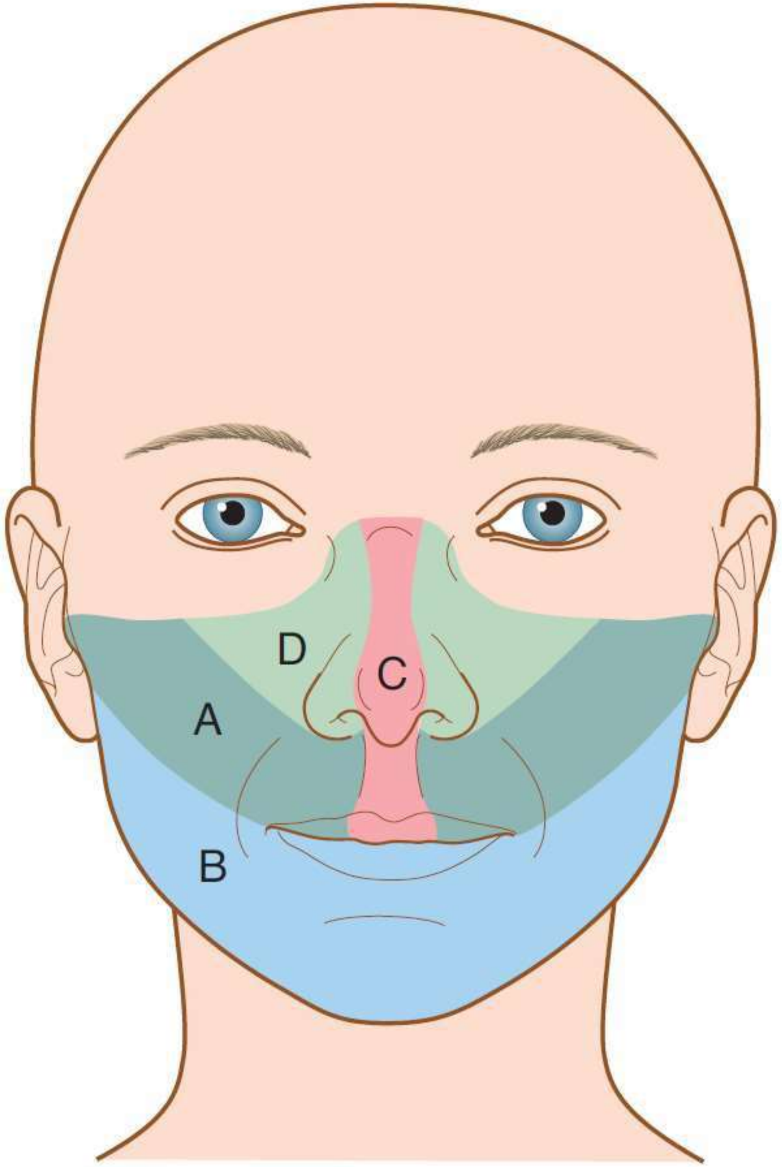



**Fig. 17.4** The contributions to the adult face from the embryonic facial processes.
A = maxillary processes;
B = mandibular processes;
C = medial nasal processes;
D = lateral nasal processes.

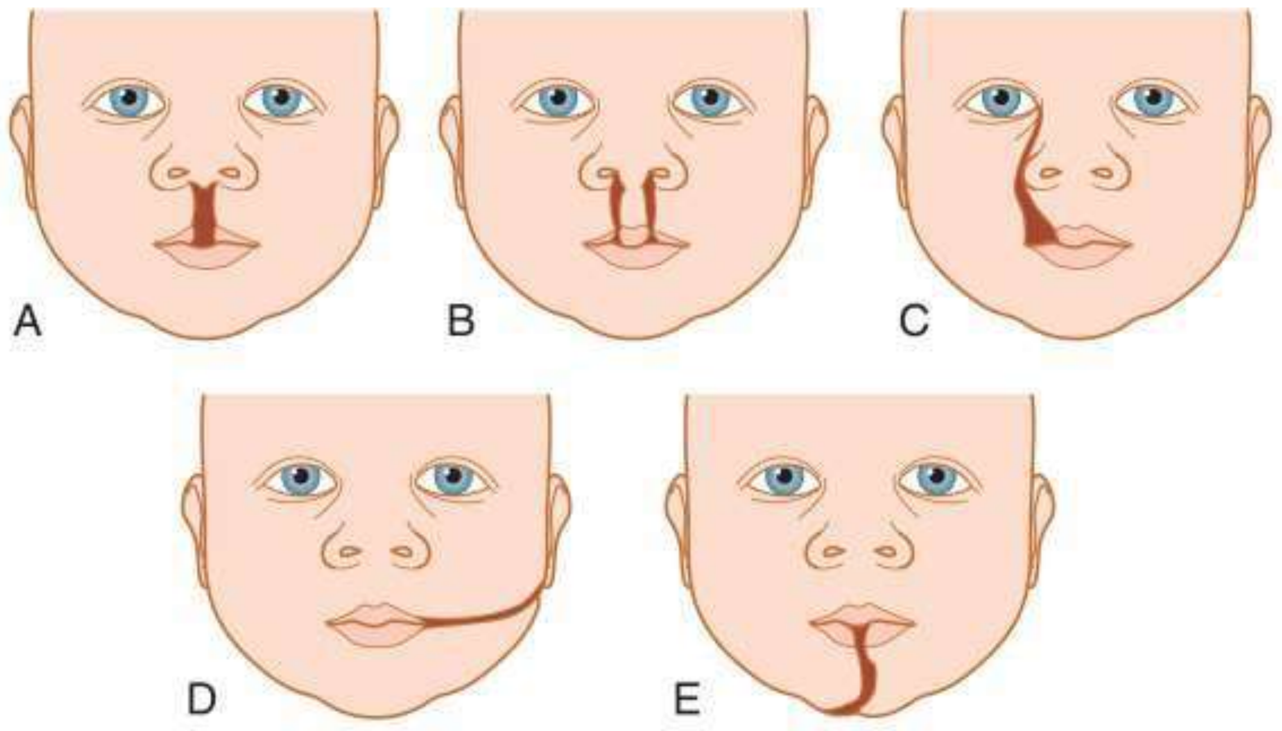



**Fig. 17.8**
Facial clefts (failure of fusion of facial processes)

A = median cleft lip;

B = bilateral cleft lip;

C = oblique facial cleft;

D = lateral facial cleft;

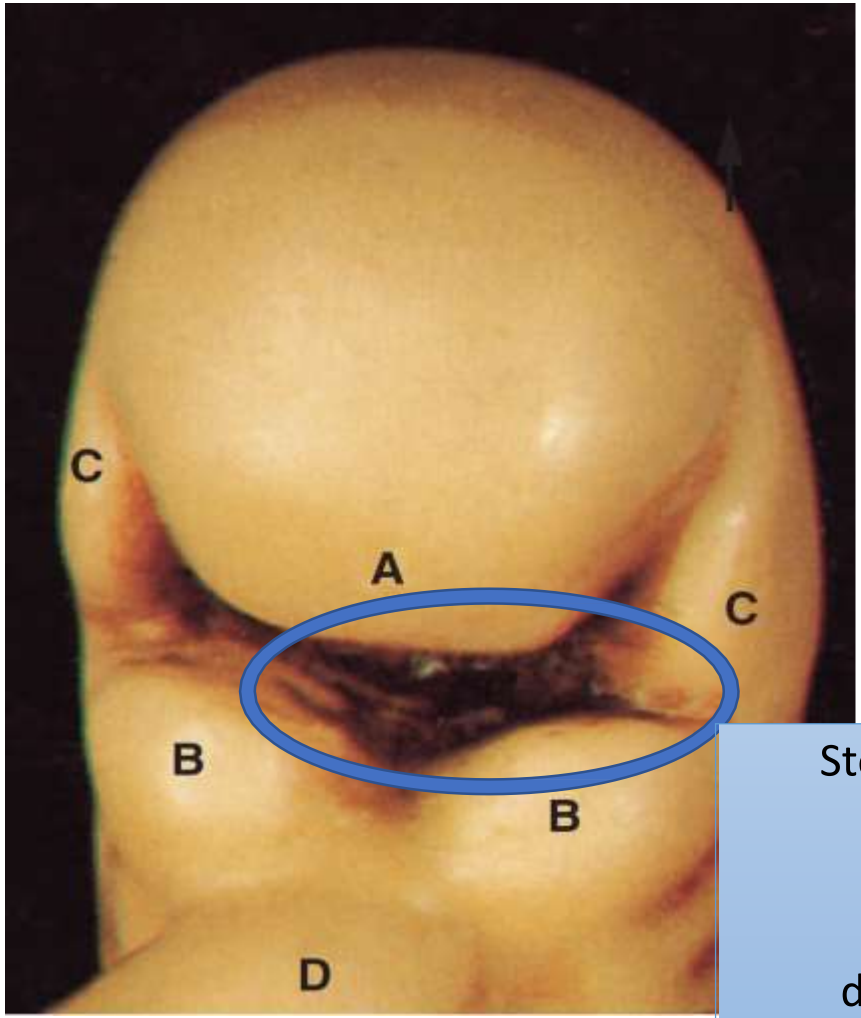
E = median mandibular cleft.

# REFERENCE/STUDY MATERIAL

# Facial Development

## During the 4th week in utero

- Primitive oral cavity (stomodeum) is bounded by five facial processes (sometimes referred to as swellings or primordia),
- And produced by proliferating zones of mesenchyme lying beneath the surface ectoderm




Stomodeum, an Ectodermal depression enlarged by disintegration of oropharyngeal membrane

**Fig. 17.1** (A) Model of a face of a 4-week-old human embryo (frontal aspect).

Frontonasal process

Mandibular processes

Maxillary processes

Pericardial swelling

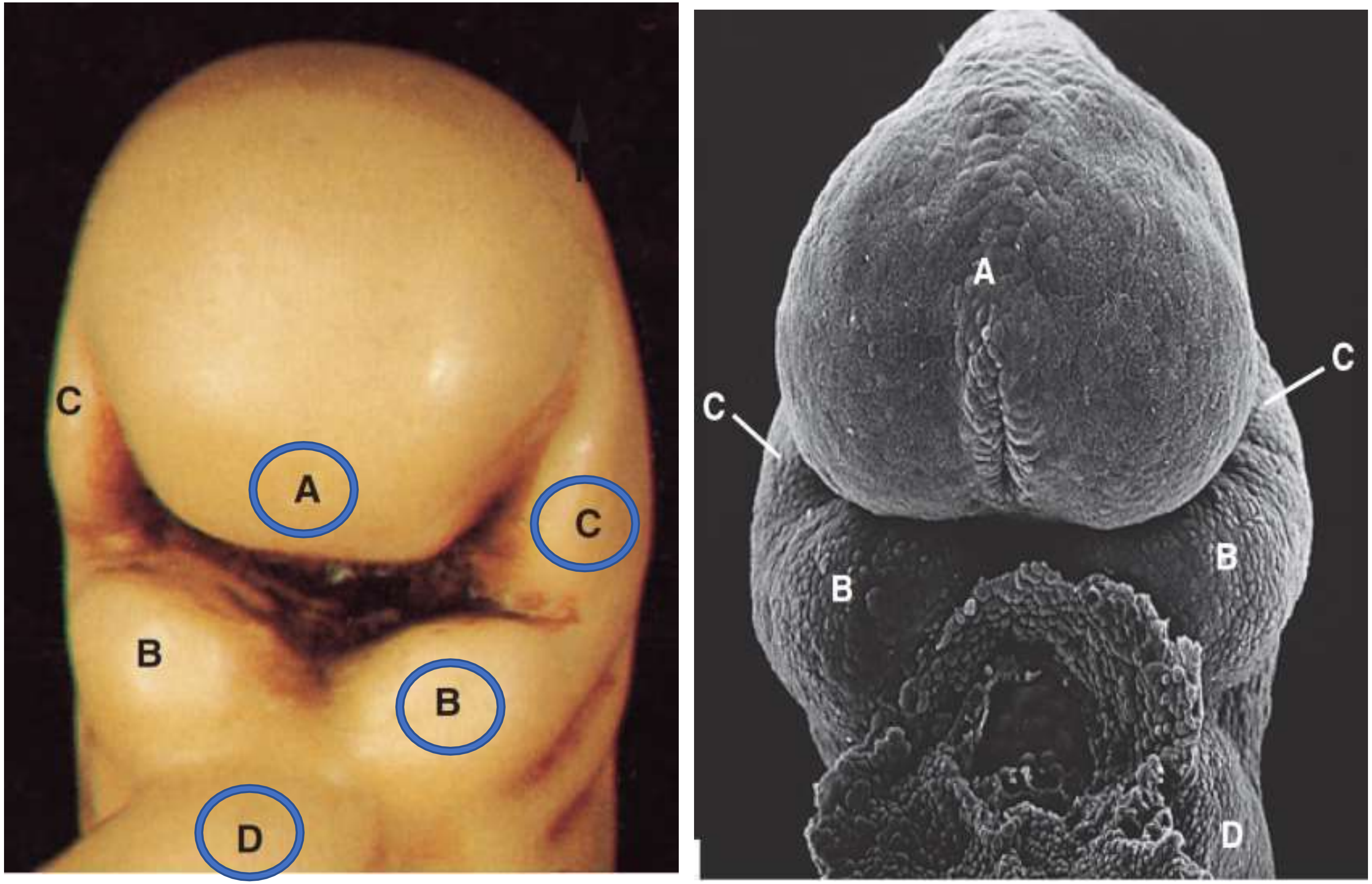
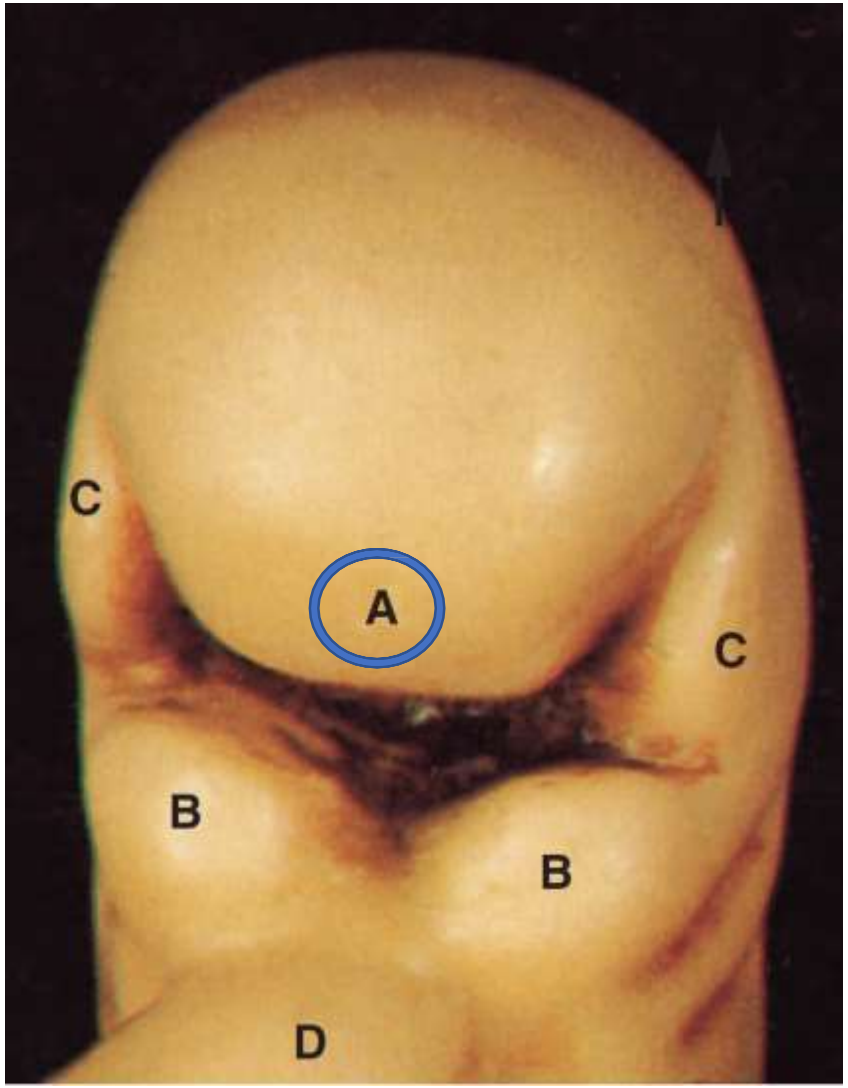



**Fig. 17.1.**(A) Model of a face of a 4-week-old human embryo (frontal aspect). (B) SEM of face

Frontonasal process




## During the 4th week in utero

- Unpaired
- formed from the mesenchyme in front of the developing forebrain

**Fig. 17.1.**(A) Model of a face of a 4-week-old human embryo (frontal aspect)

# Craniofacial Development

**Face develops in human between 4th and 10th weeks of intra-uterine life.**

**Craniofacial development is related to derivatives of the pharyngeal (branchial) arches**

**It occurs by epithelial–mesenchymal interactions with coordination of specific genes and signalling molecules.**

**Craniofacial anomalies are amongst the most common congenital anomalies**
**found in man.**

# Mechanisms involved in Craniofacial Development

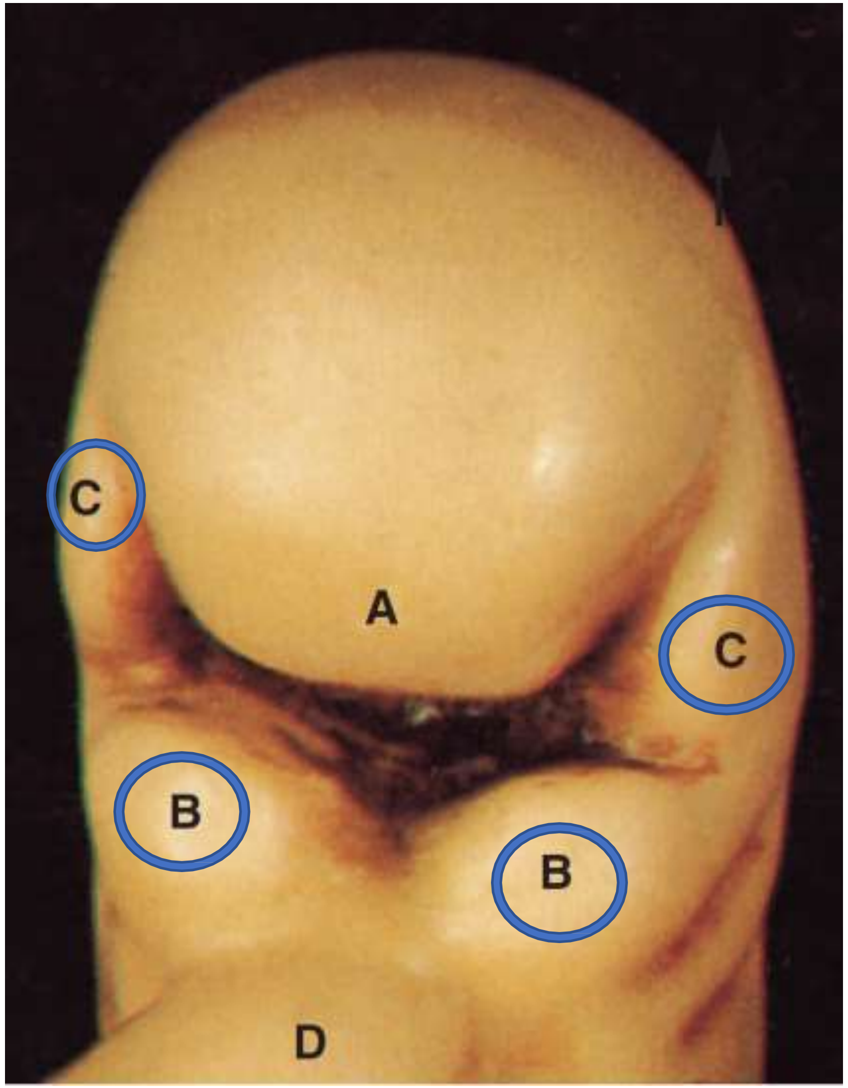
**Merging of mesenchymal facial processes**

**Intramembranous calcification around the cartilage of first pharyngeal arch to form the body of the mandible,**

**Hydration of palatal shelves to produce shelf elevation for the formation of definitive palate.**

Mandibular processes

Maxillary processes




## During the 4th week in utero

- Paired
- derived from the first pharyngeal (branchial) arches.

**Fig. 17.1.**(A) Model of a face of a 4-week-old human embryo (frontal aspect)

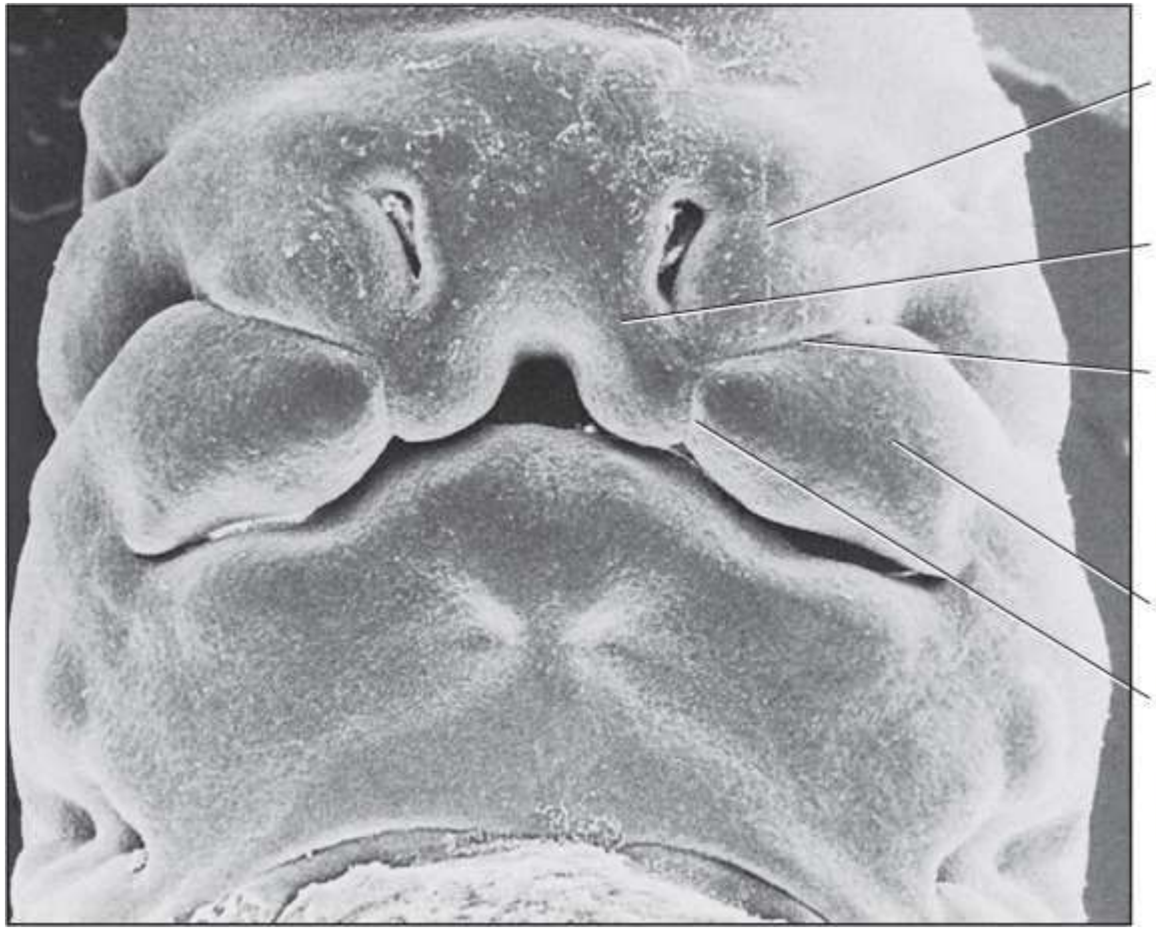



**FIGURE 3-14** Scanning electron micrograph of a human embryo at around 6 weeks of development

## Facial Processes

- demarcated by grooves
- that become flattened
- out by the proliferative
- and migratory activity
- of the underlying
- mesenchyme.

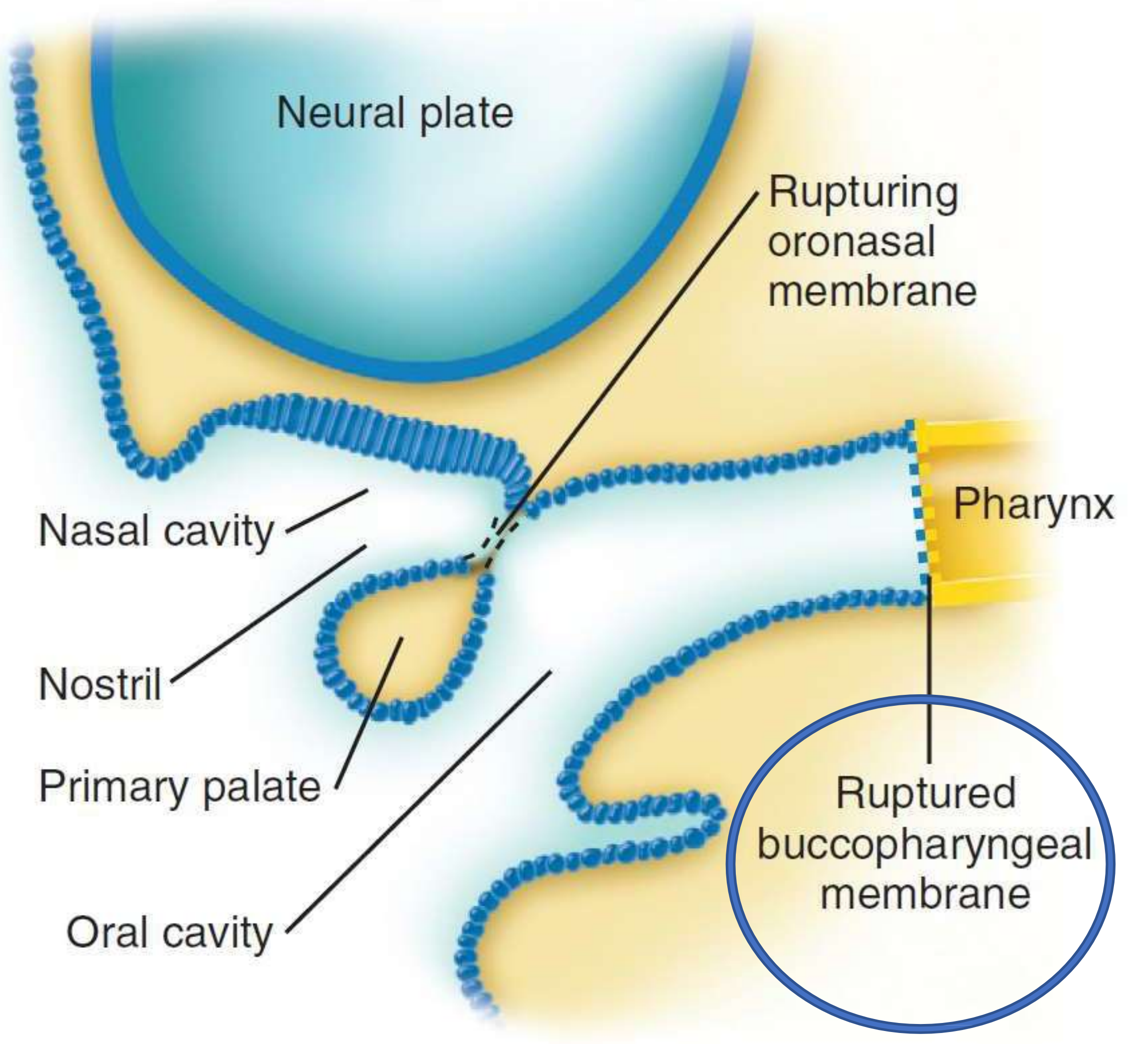



**FIGURE 3-23** Summary of the development of the oral cavity as seen in midsagittal section.

## Oropharyngeal membrane

- separates the primitive oral cavity from the developing pharynx.
- Bilaminar
- Composed of an outer ectodermal layer and an inner endodermal layer.
- Breaks down to establish continuity between the ectoderm-lined oral cavity and the endoderm-lined pharynx.
- Although not detectable in the adult
- Demarcation zone between mucosa derived from ectoderm and endoderm correspond to a region lying just behind the permanent third molar tooth.

# Oropharyngeal Membrane

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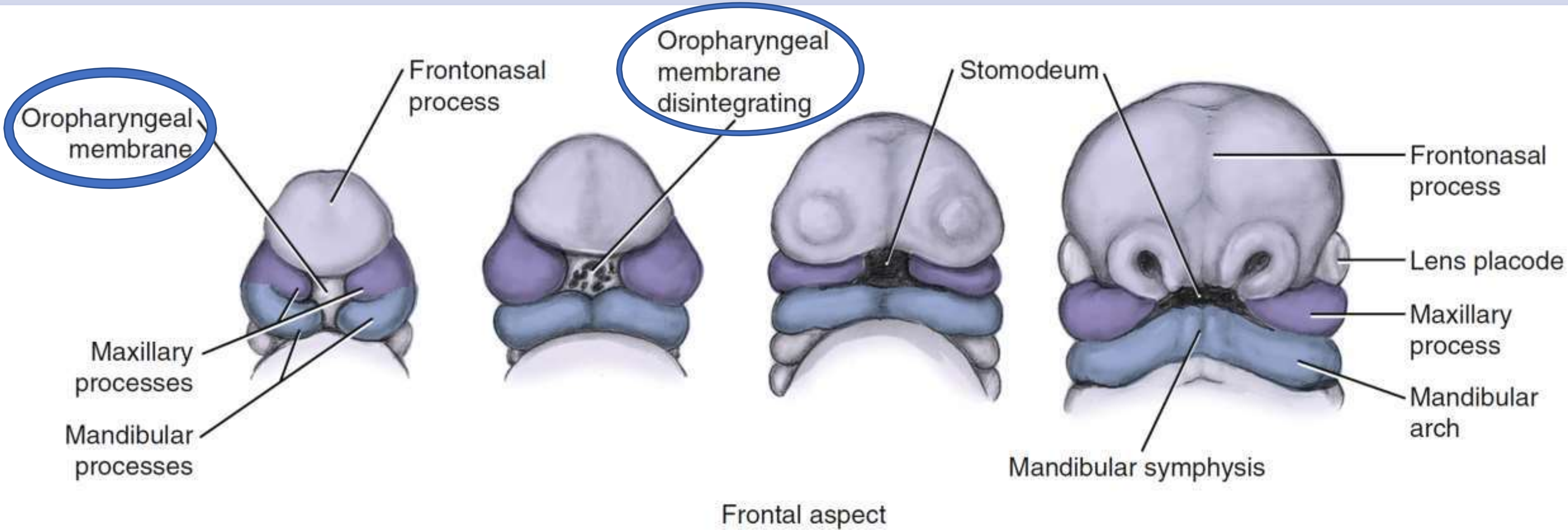
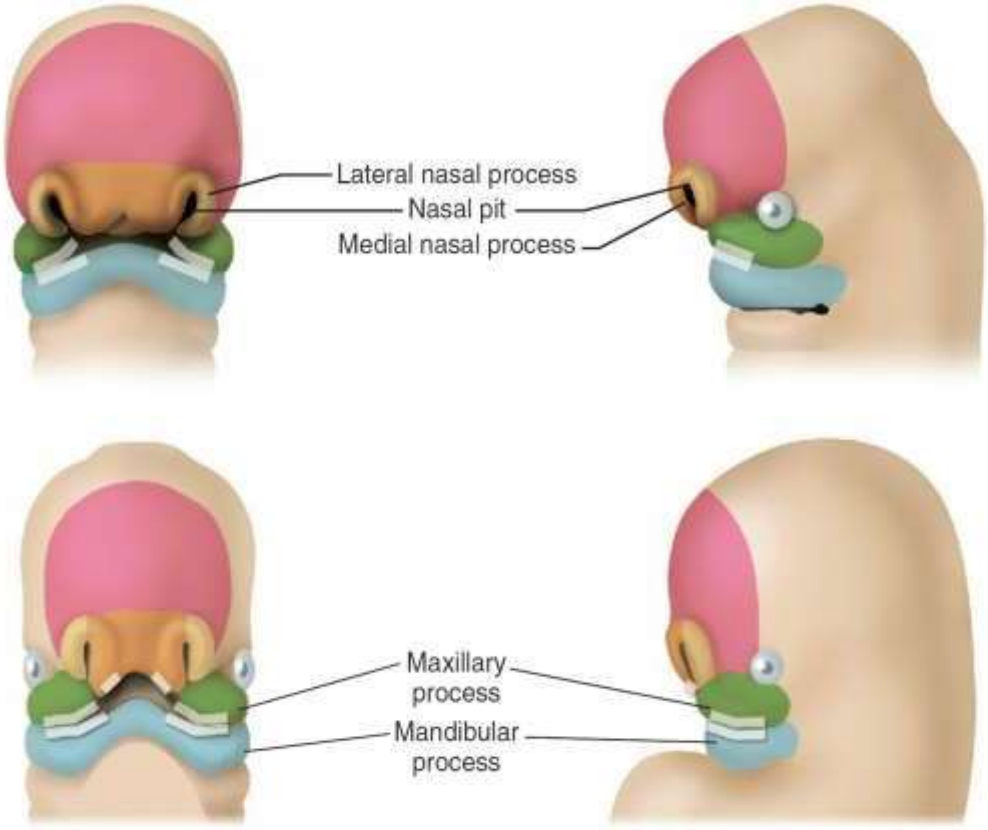


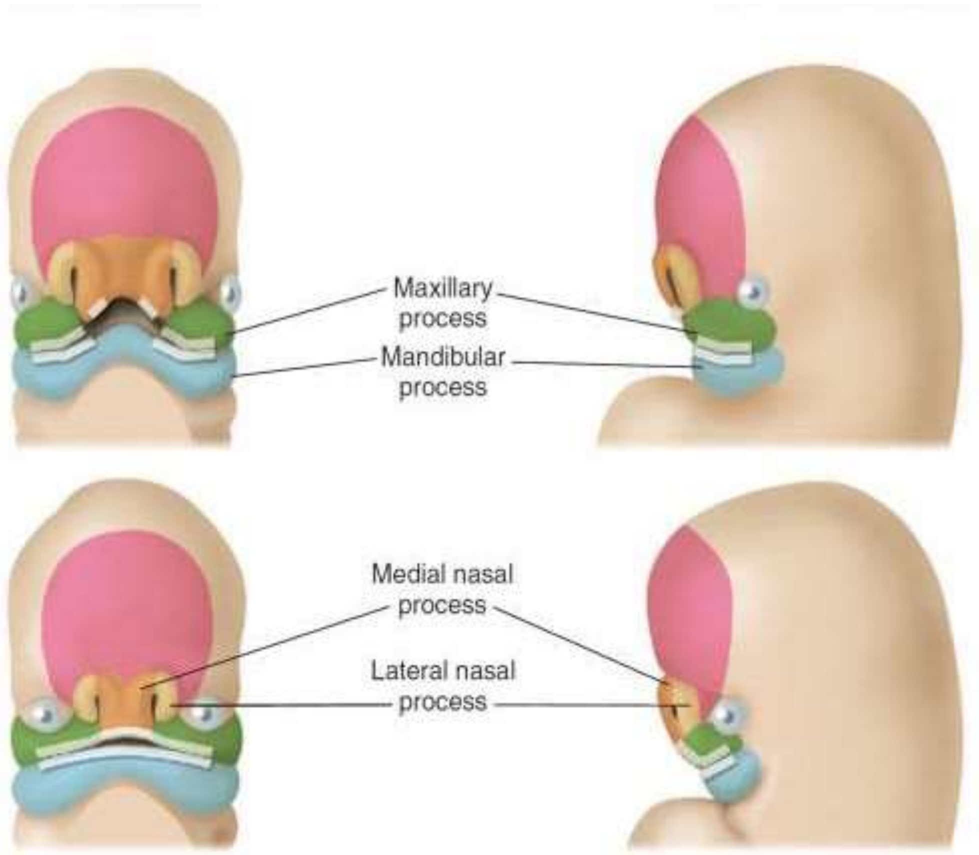
During the 5th week in utero

- localised thickenings of ectoderm give rise to the nasal and optic placodes.
- These placodes will form the olfactory epithelium and the lenses of the eyes.
- The nasal placodes sink into the underlying mesenchyme, forming two blind-ended nasal pits (the primitive nasal cavities).



During the 5th week in utero

- Proliferation of mesenchyme from the frontonasal process around the openings of the nasal pits produces the medial and lateral nasal processes.
- maxillary processes enlarge and grow forwards and medially.



## During the 5th week in utero

- A = optic placode;
- B = nasal pit;
- C = medial nasal process;
- D = lateral nasal process;
- E = maxillary process;
- F = mandibular process;
- G = second pharyngeal arch.

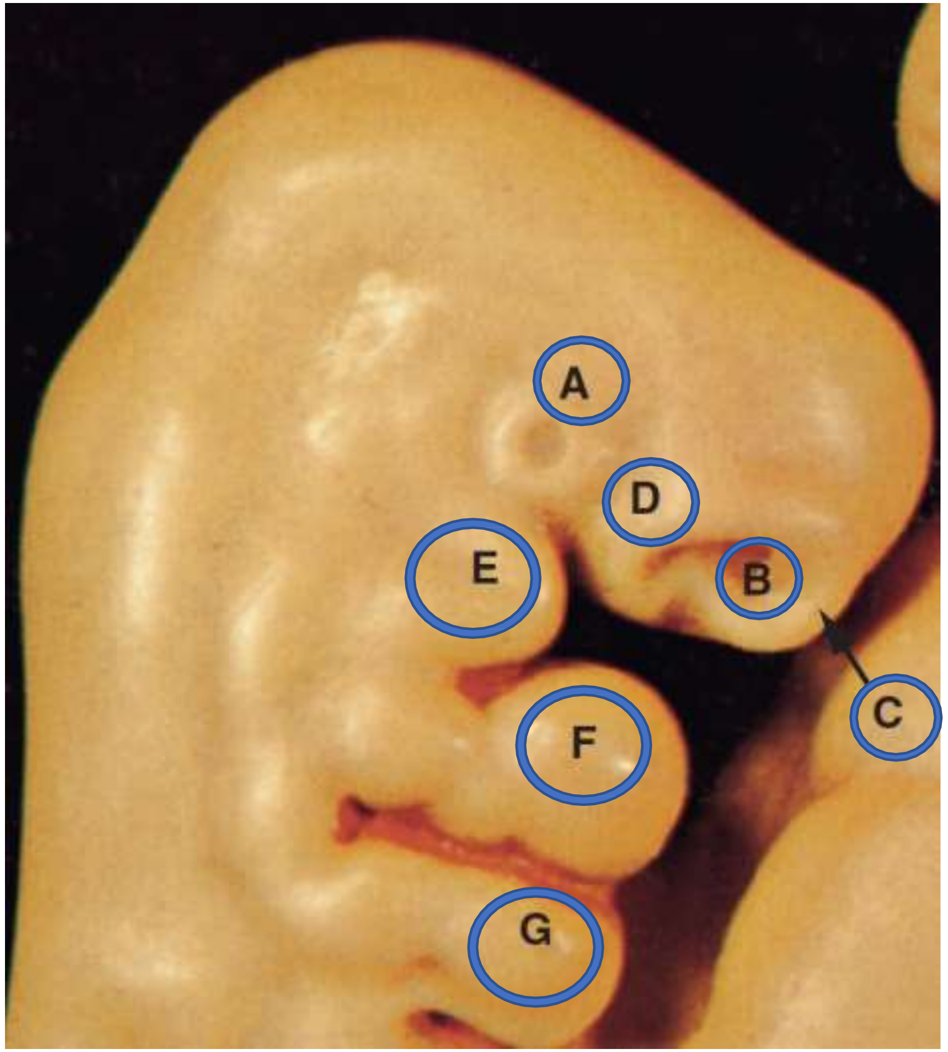
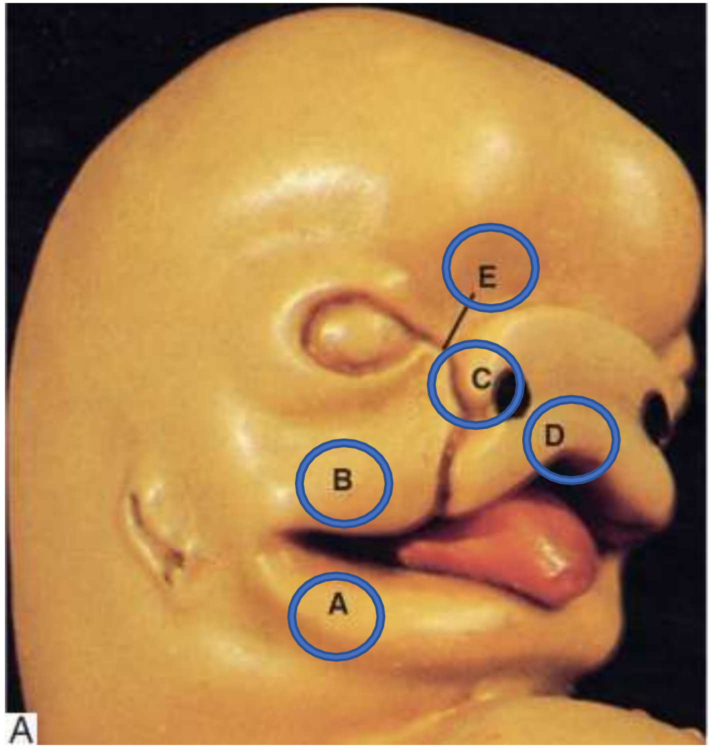



**Fig. 17.2**

# 6-week-old embryo

- two mandibular processes fuse in the midline to form lower jaw.
- mandibular and maxillary processes are continuous at the angle of the mouth.



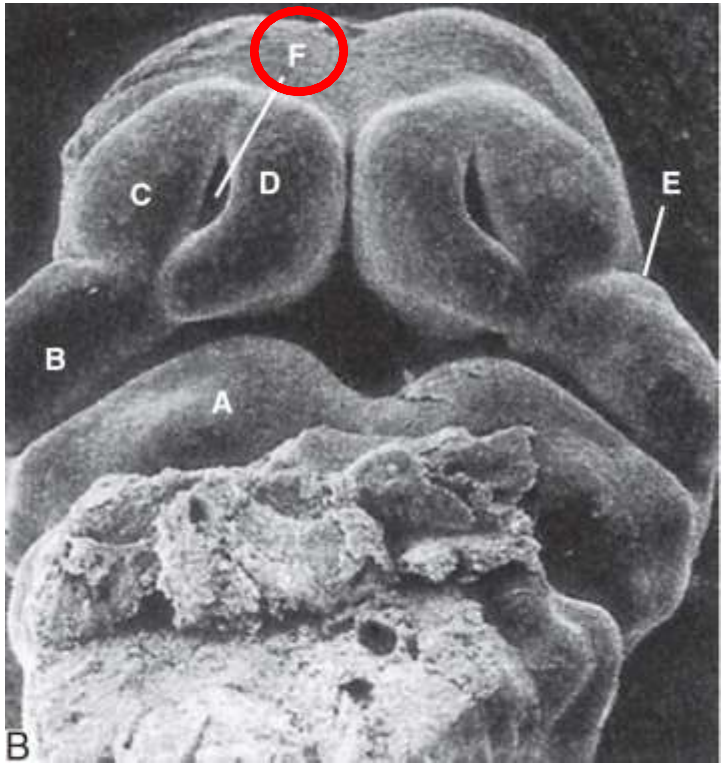



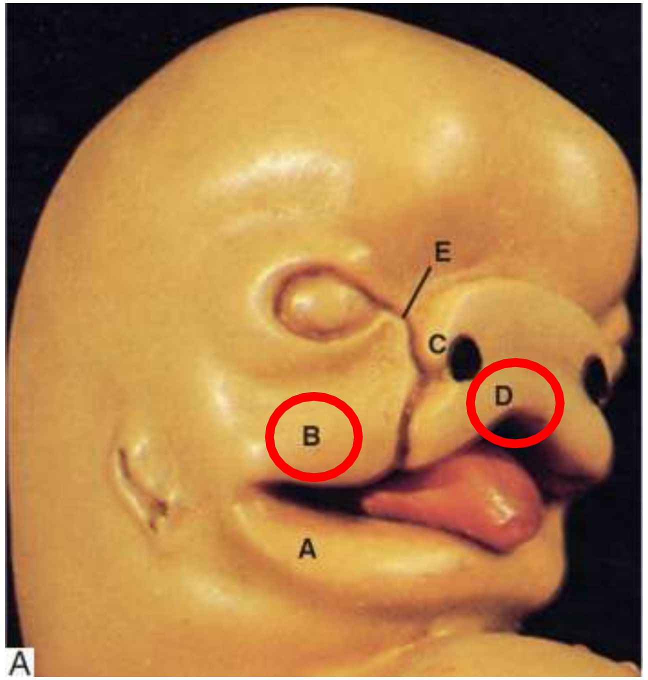

**Fig. 17.3** (A) Model of a face of a 6-week-old human embryo (lateral aspect). (B) SEM of Face
A = mandibular process; B = maxillary process;
C = lateral nasal process; D = medial nasal process;
E = naso-optic furrow ; F = nasal pit

## 6-week-old embryo

- Below the lateral nasal processes

❿maxillary processes meet the medial nasal processes by the overgrowth or mesenchyme invasion, for forming upper lip.

❿Another view is that the middle third of the upper lip (the intermaxillary segment)(premaxilla)(primary palate) being therefore derived from the merged medial nasal processes of the frontonasal process.



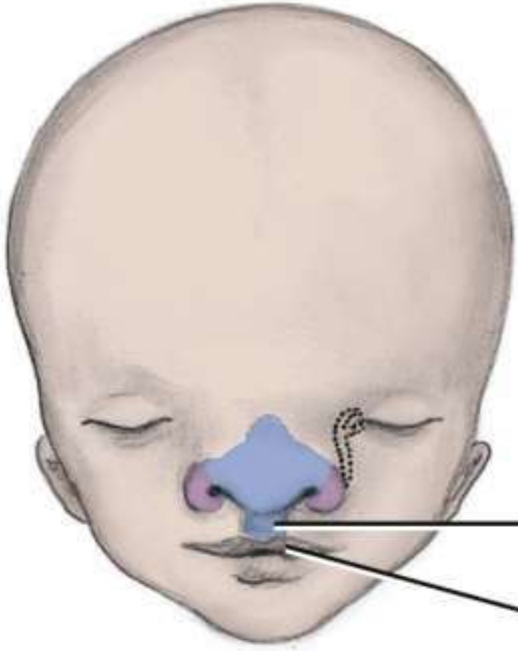
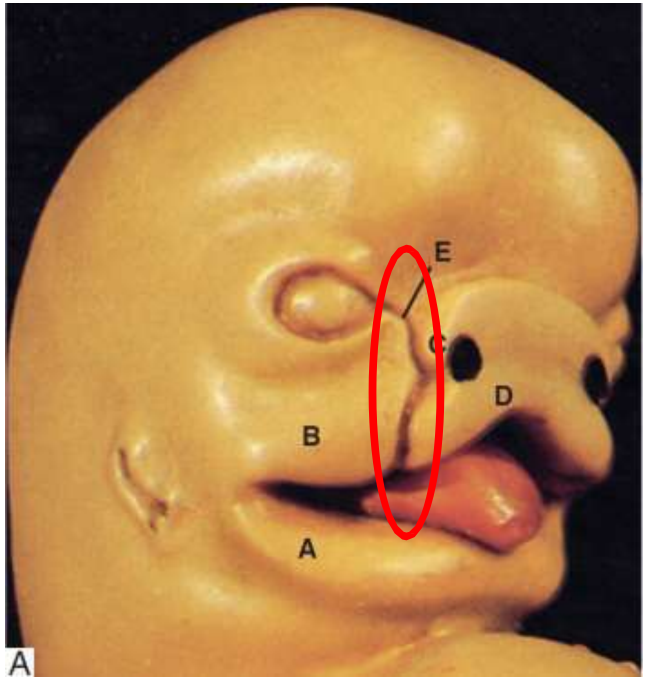




**Fig. 17.3** (A) Model of a face of a 6-week-old human embryo (lateral aspect).
B = maxillary process;
D = medial nasal process;

## 6-week-old embryo

- Between the merging maxillary and lateral nasal processes lie naso-optic furrows (alternatively termed as nasolacrimal grooves).
- From each furrow, a solid ectodermal rod of cells sinks below the surface and canalises to form the nasolacrimal duct.



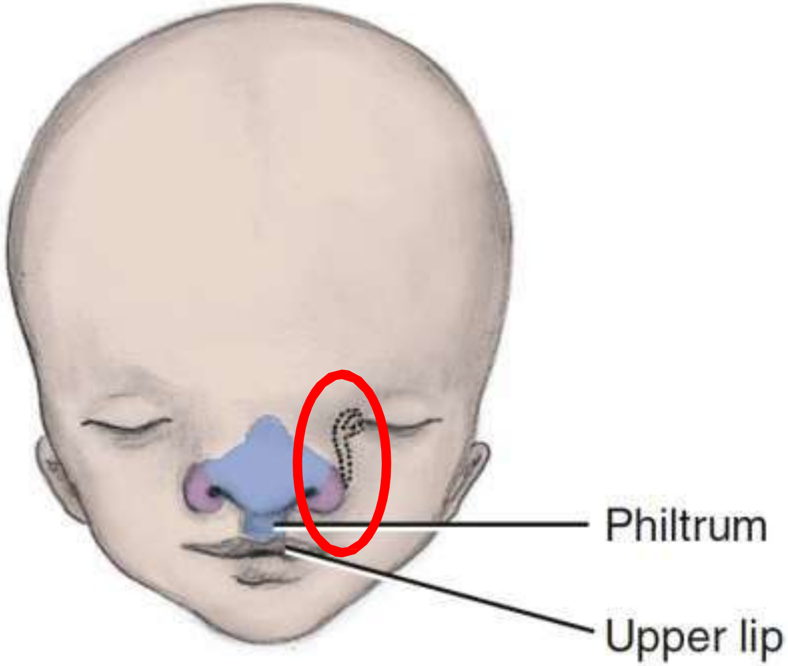
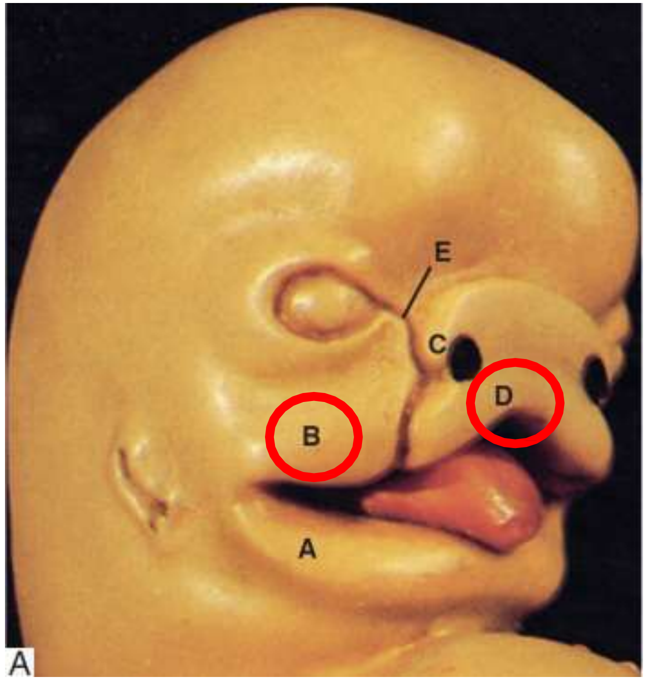




**Fig. 17.3** (A) Model of a face of a 6-week-old human embryo (lateral aspect).
B = maxillary process;
D = medial nasal process;

## Facial placods

- Below the lateral nasal processes
- ➓ maxillary processes meet the medial nasal processes by the overgrowth or mesenchyme invasion, for forming upper lip.
- ➓ Another view is that the middle third of the upper lip (the intermaxillary segment)(premaxilla)(primary palate) being therefore derived from the merged medial nasal processes of the frontonasal process.



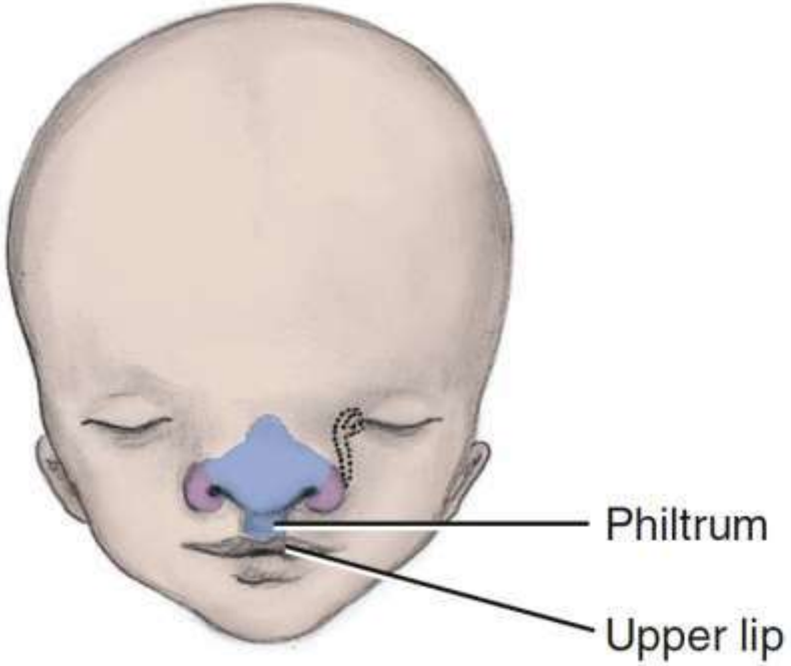




**Fig. 17.3** (A) Model of a face of a 6-week-old human embryo (lateral aspect).
B = maxillary process;
D = medial nasal process;

Facial Placodes These

sensory placodes are thickenings of the surface ectoderm that initially arise from a common ‘preplacodal field’ that lies close to the anterior neural crest border.

1. Nasal placodes

These are situated towards the front of the developing head (on either side of the frontonasal process) and give rise to the olfactory epithelia

2. Lens placodes

These invaginate to form the lens vesicles (the lenses of the eyes).

3. Otic placodes;

These appear over the regions of the developing hindbrain to form the otic vesicles (internal ears; the primordium of the

Table 17.1 Signaling factors associated with lens, nasal and otic placodes

| Signalling factor family | Lens placode | Nasal placode | Otic placode |
|---|---|---|---|
| FGF | FGFR | FGF | FGFR and FGF |
| PDGF | – | – | – |
| RA | – | – | – |
| Shh | – | – | – |
| TGF-β | BMP4 and BMP7 | – | – |
| Wnt | – | – | Wnt |

# DEVELOPMENT OF NASAL CAVITIES

The nasal placodes invaginate from the surface by combination of active growth of its epithelium and by proliferation of the mesenchyme at the edges of the placode

The nasal placodes do not become completely internalised and thus form two blind-ended pits, the nasal pits (nasal sac).

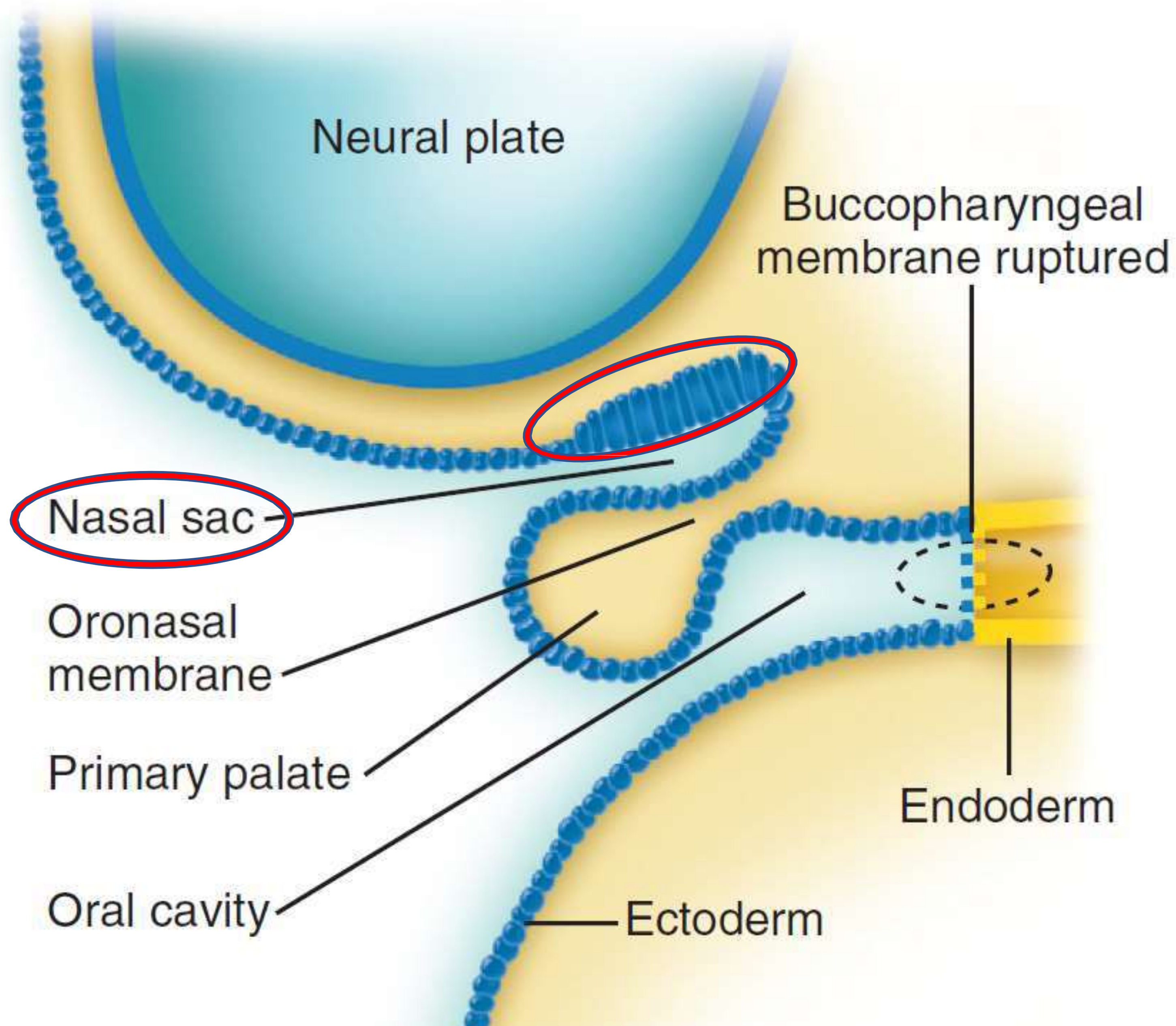
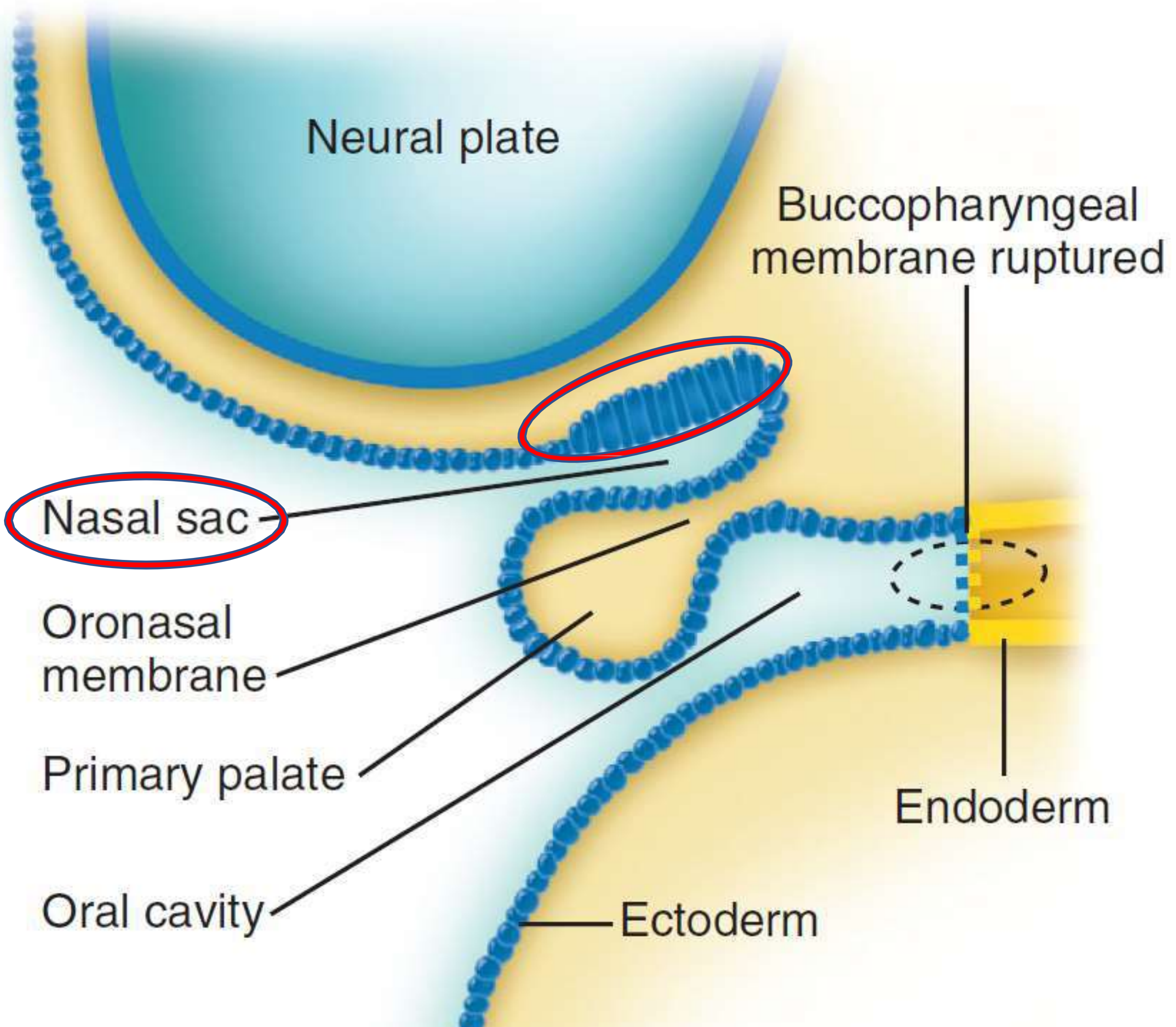



**FIGURE 3-23** Summary of the development of the oral cavity as seen in midsagittal section.

# DEVELOPMENT OF NASAL CAVITIES

- ✓ Nasal placodes are separated in the midline by the primary nasal septum.
- ✓ Nasal pits continue to deepen and the nasal placodes come to lie in the roof of the nasal pits

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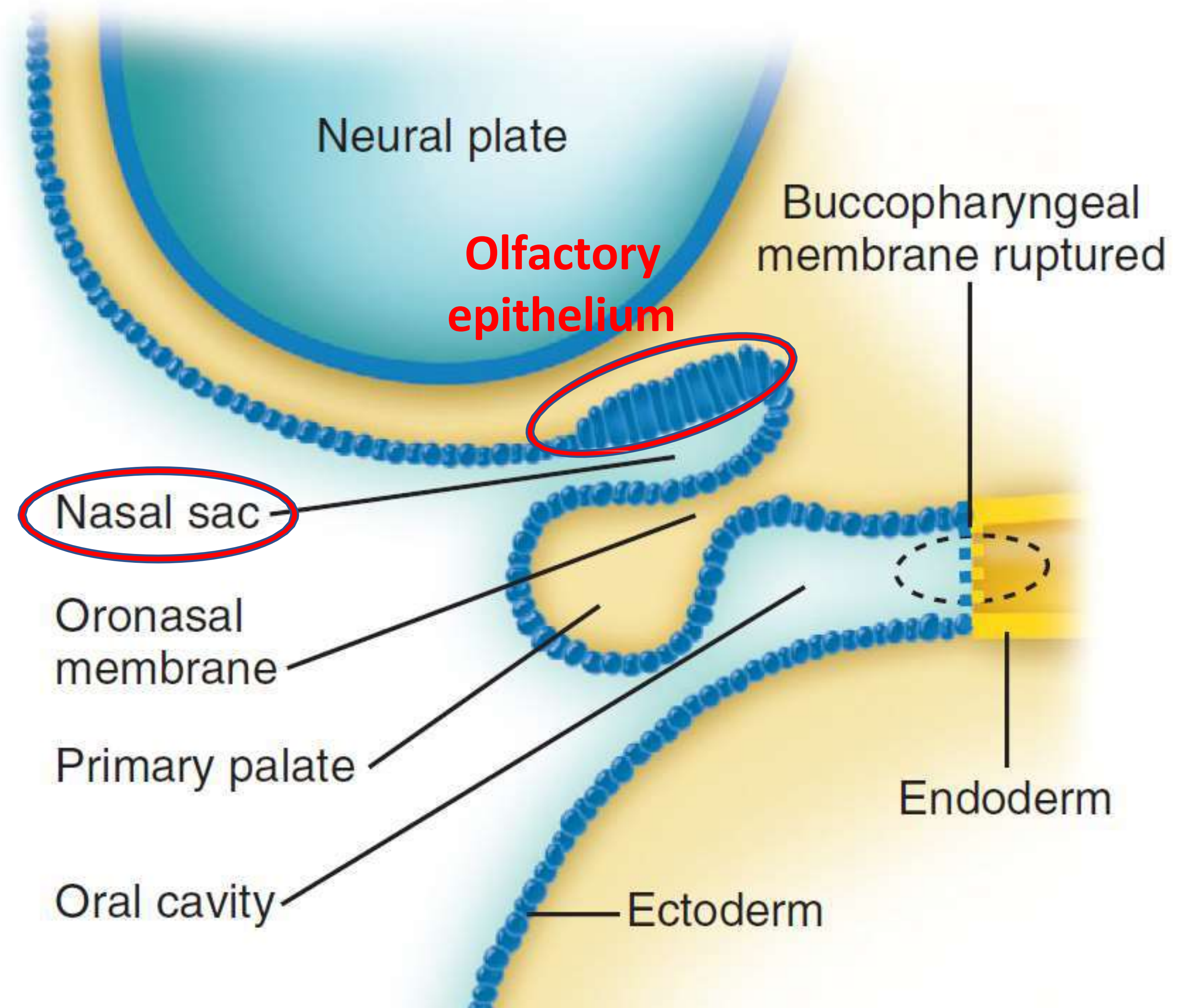
# DEVELOPMENT OF NASAL CAVITIES

- ✓ Nasal placodes are separated in the midline by the primary nasal septum.
- ✓ Nasal pits continue to deepen and the nasal placodes come to lie in the roof of the nasal pits to form the olfactory epithelium

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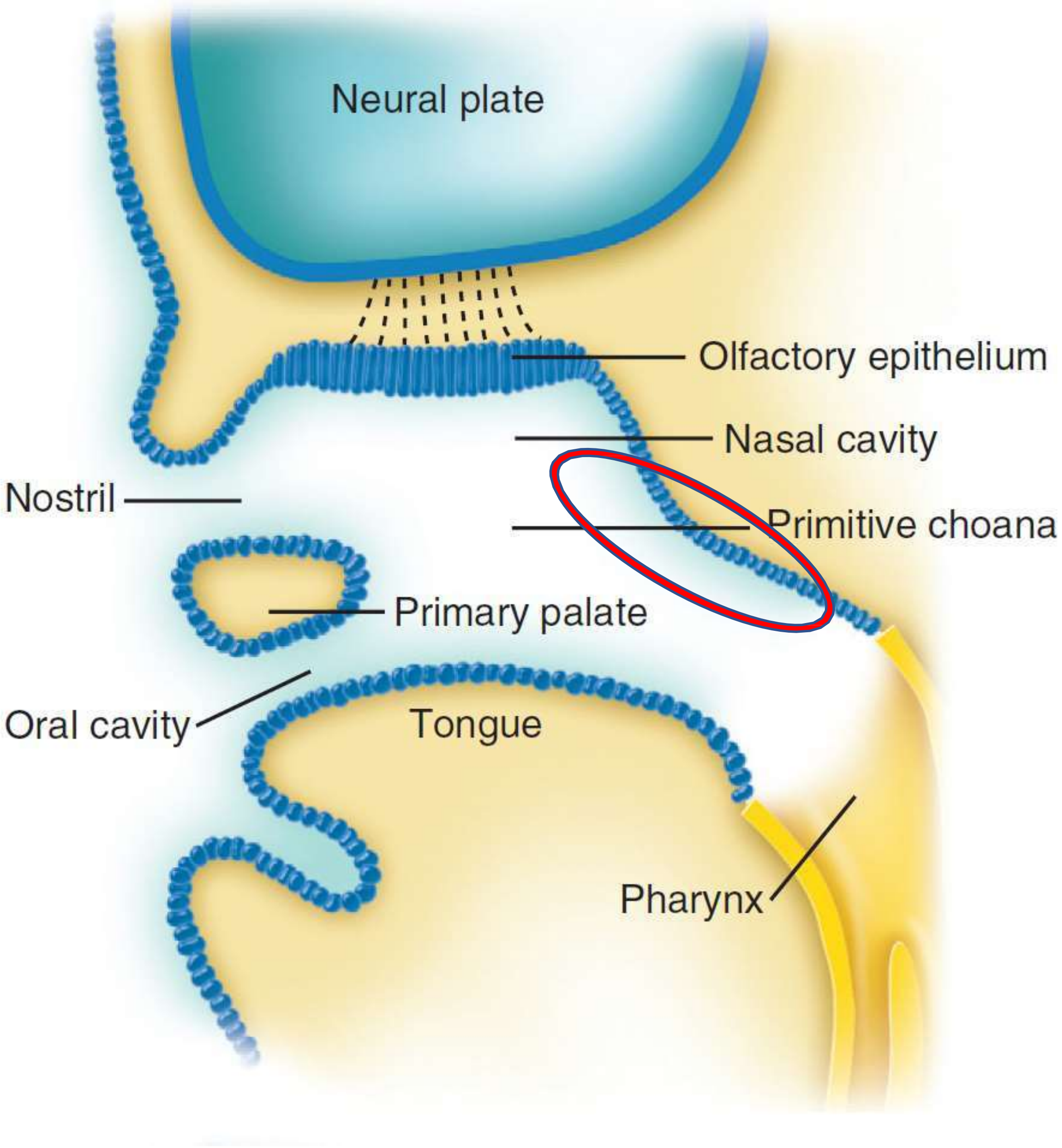
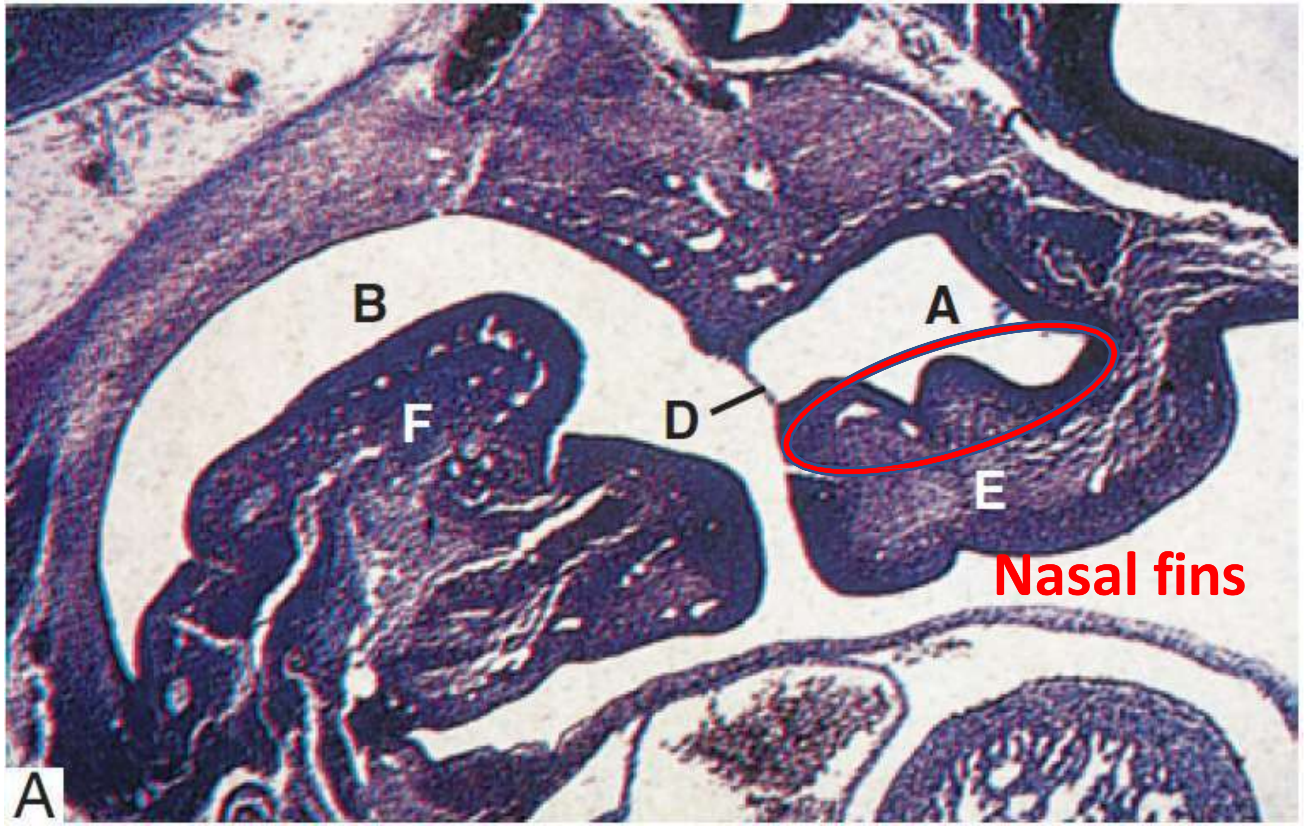
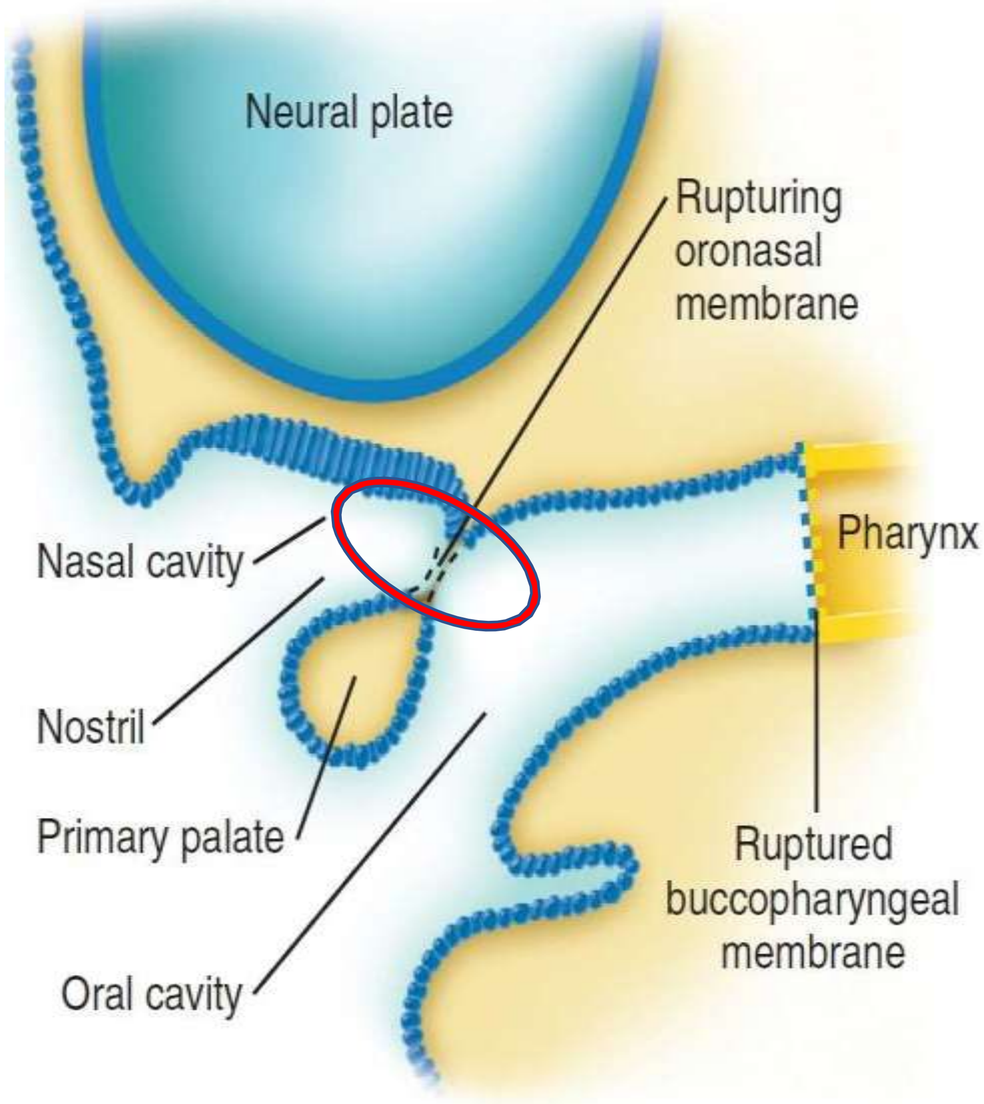
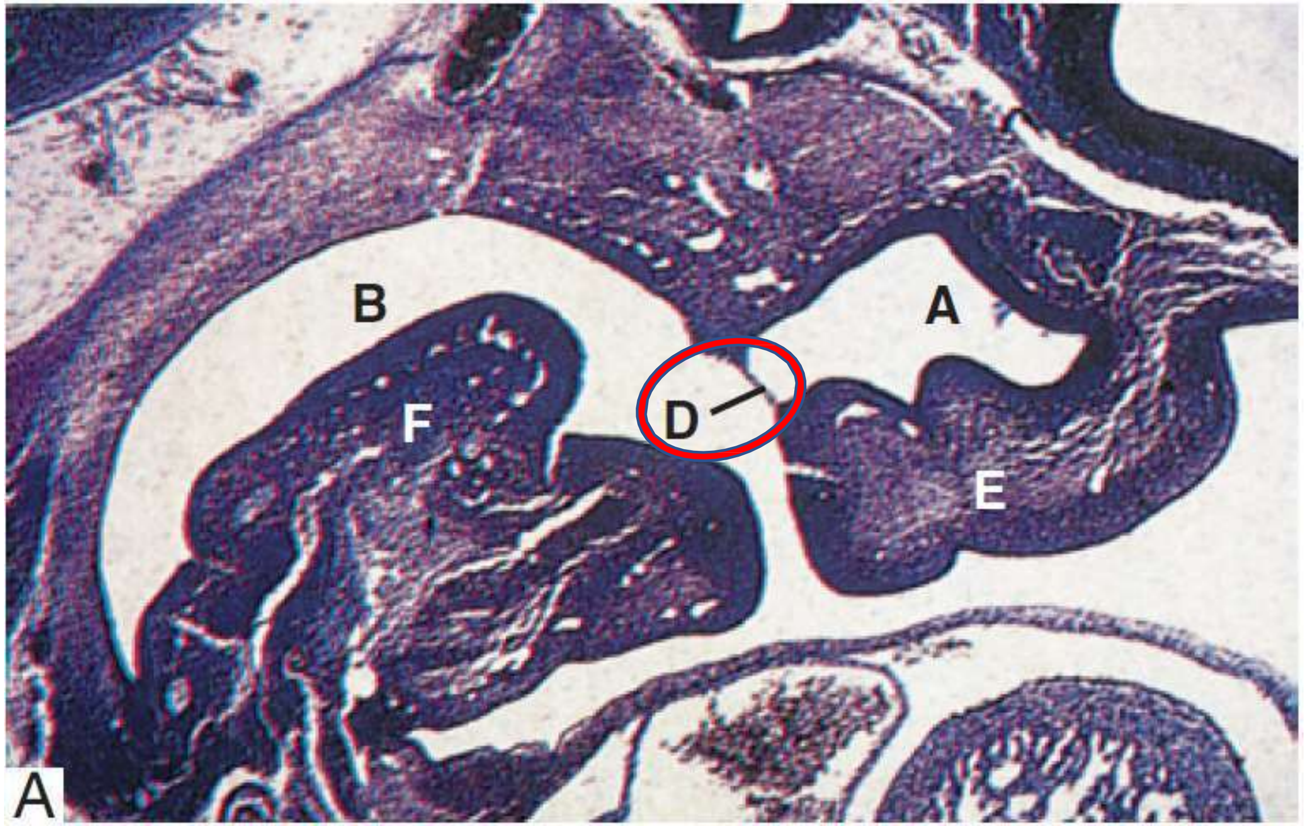



Nasal pits continue to extend into the developing head of the embryo until the roof of the primitive oral cavity is partitioned from the floor of each nasal pit by a sheet of epithelium termed the nasal fin.




Sagital section of developing embryo; Tencat




- ✓ Nasal fin thins to form an oronasal membrane separating the nasal pit from the developing oral cavity.
- ✓ By the end of the 5th week, the oronasal membrane ruptures to produce communication between oral and nasal cavities.

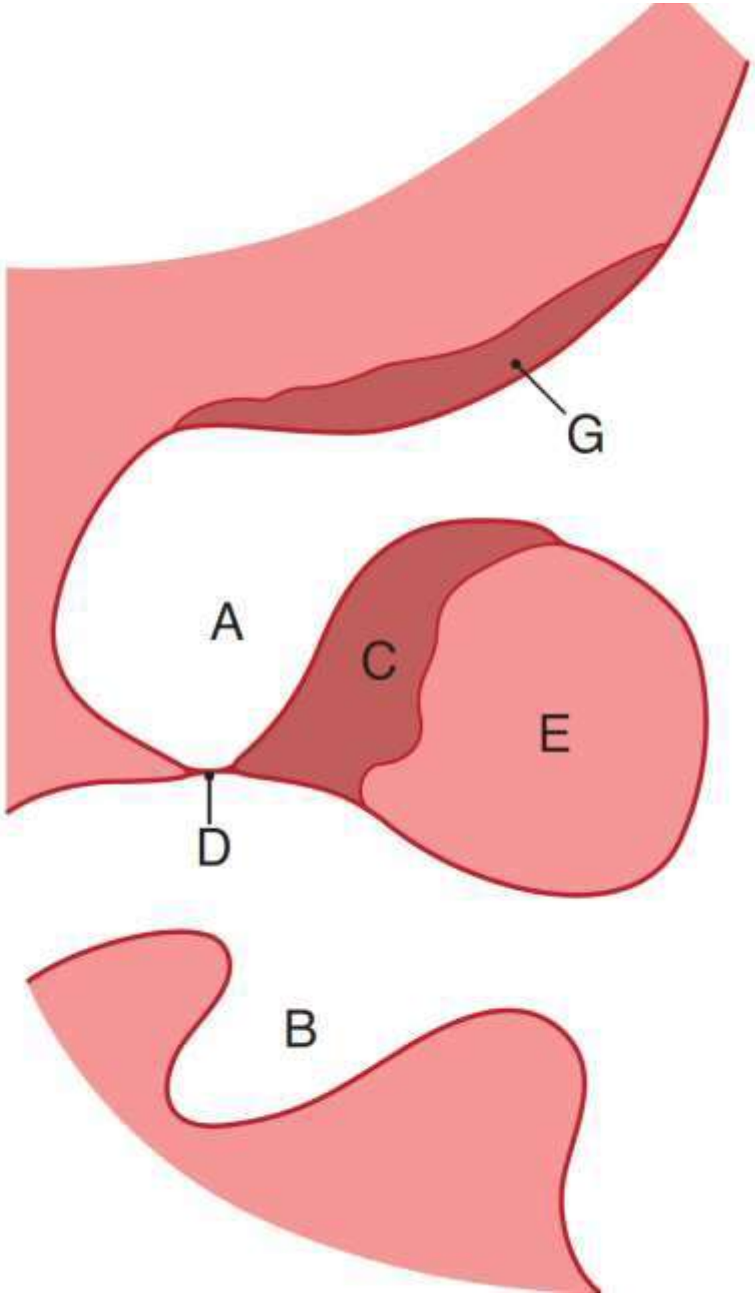
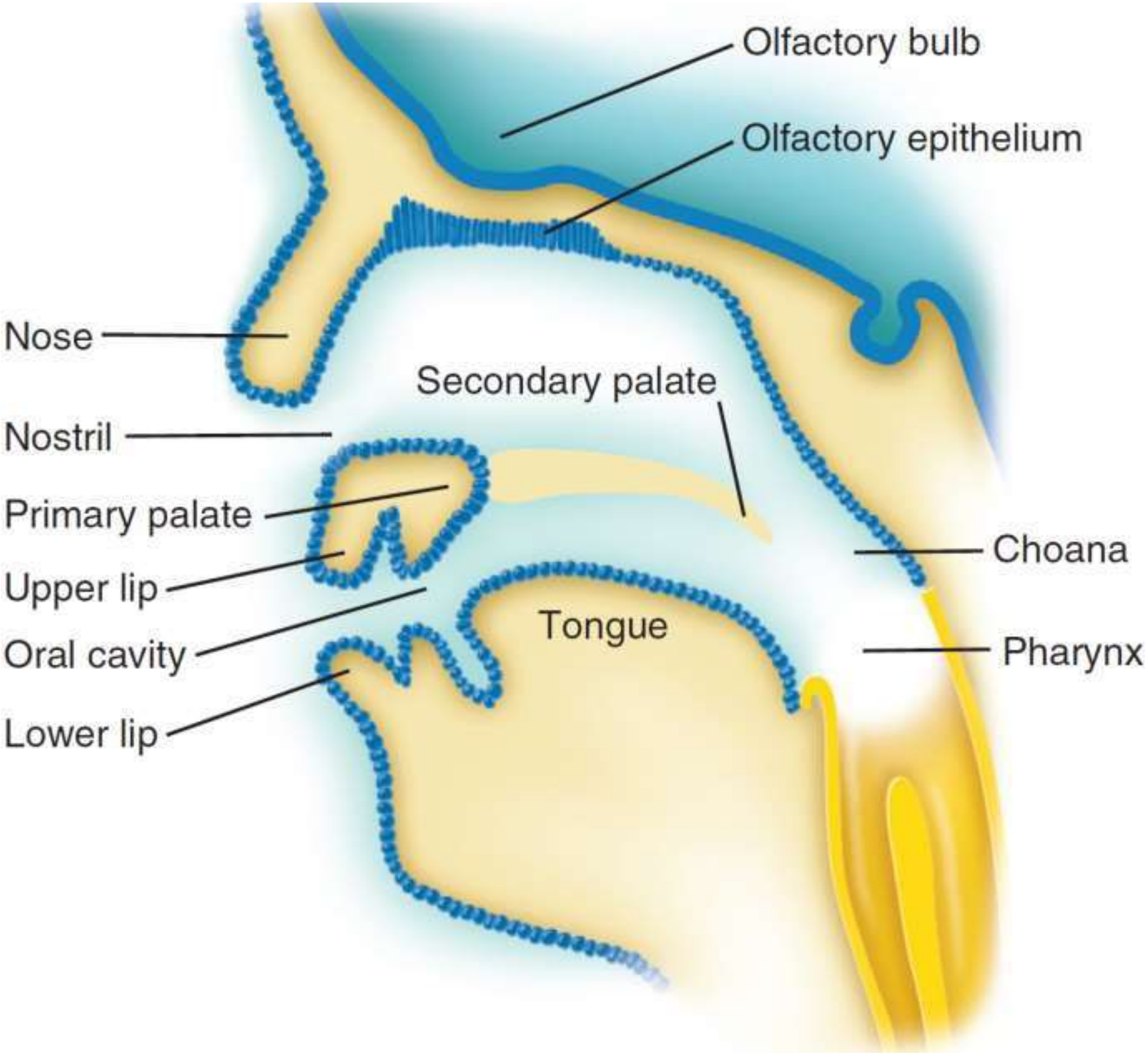



**Fig. 17.6** Sagittal sections of the head illustrating aspects of the development of the nasal cavity.
A = nasal pit;
B = oral cavity;
C = nasal fin;
D = oronasal membrane;
E = maxillary isthmus;
F = developing tongue;
G = nasal placode in root of nasal pit

- ✓ The anterior openings of the primitive nasal cavities become filled with epithelial cells to form ‘epithelial plugs’. Eventually, these plugs disintegrate so that the nasal passages become open again.
- ✓ The remains of the plugs contribute to the formation of the vestibule of the nose up to the ridge (the limen nasi) that marks the start of the nasal cavity proper.




Sagital section of developing embryo; Tencat

Thank you for attention